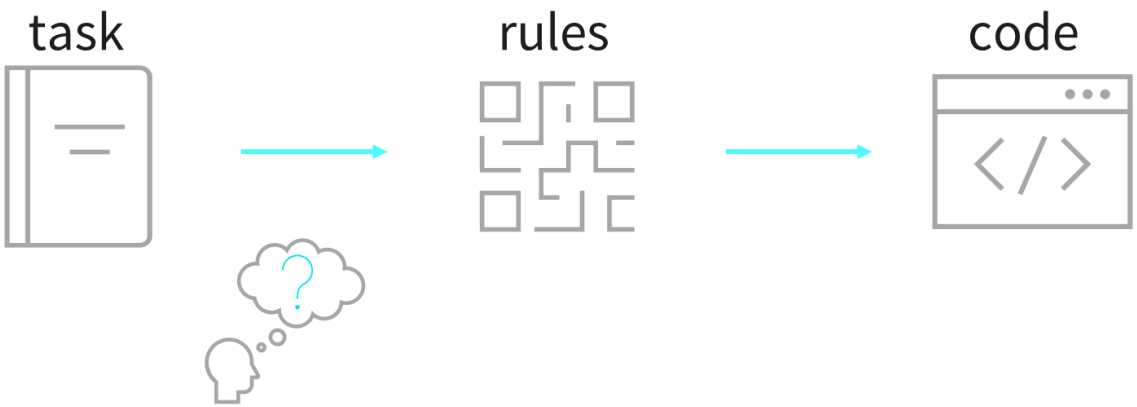
Machine Learning

Holberton

Why Machine Learning

Why Machine Learning

- For some problems we cannot simply write the rules



– What if computer systems could learn through trial and error?

Why Machine Learning

Social media features

Sentiment analysis

Image recognition

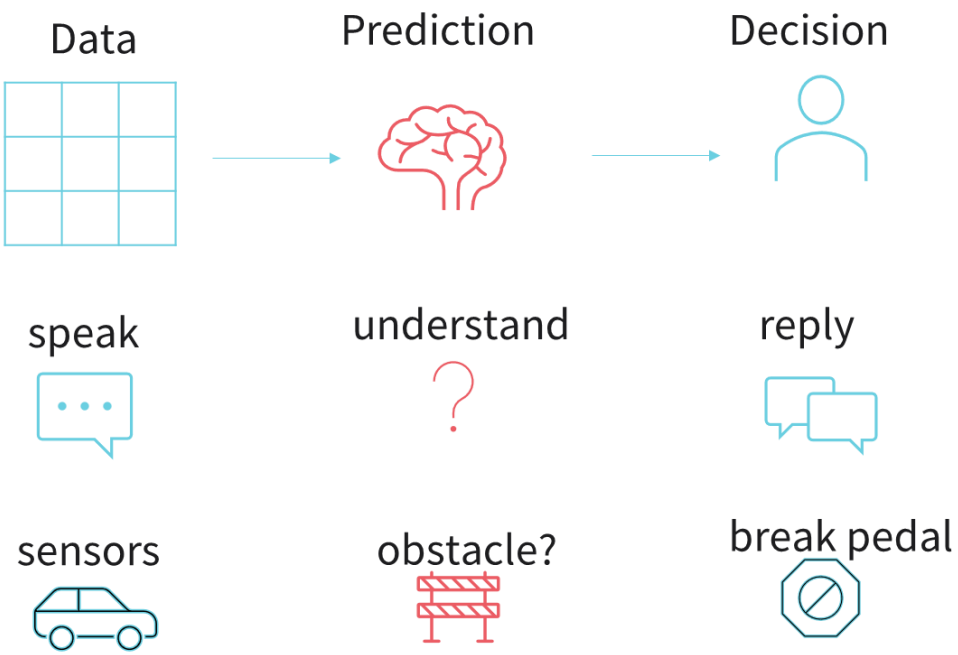
Product recommendations

Medical analysis

Stock market prediction

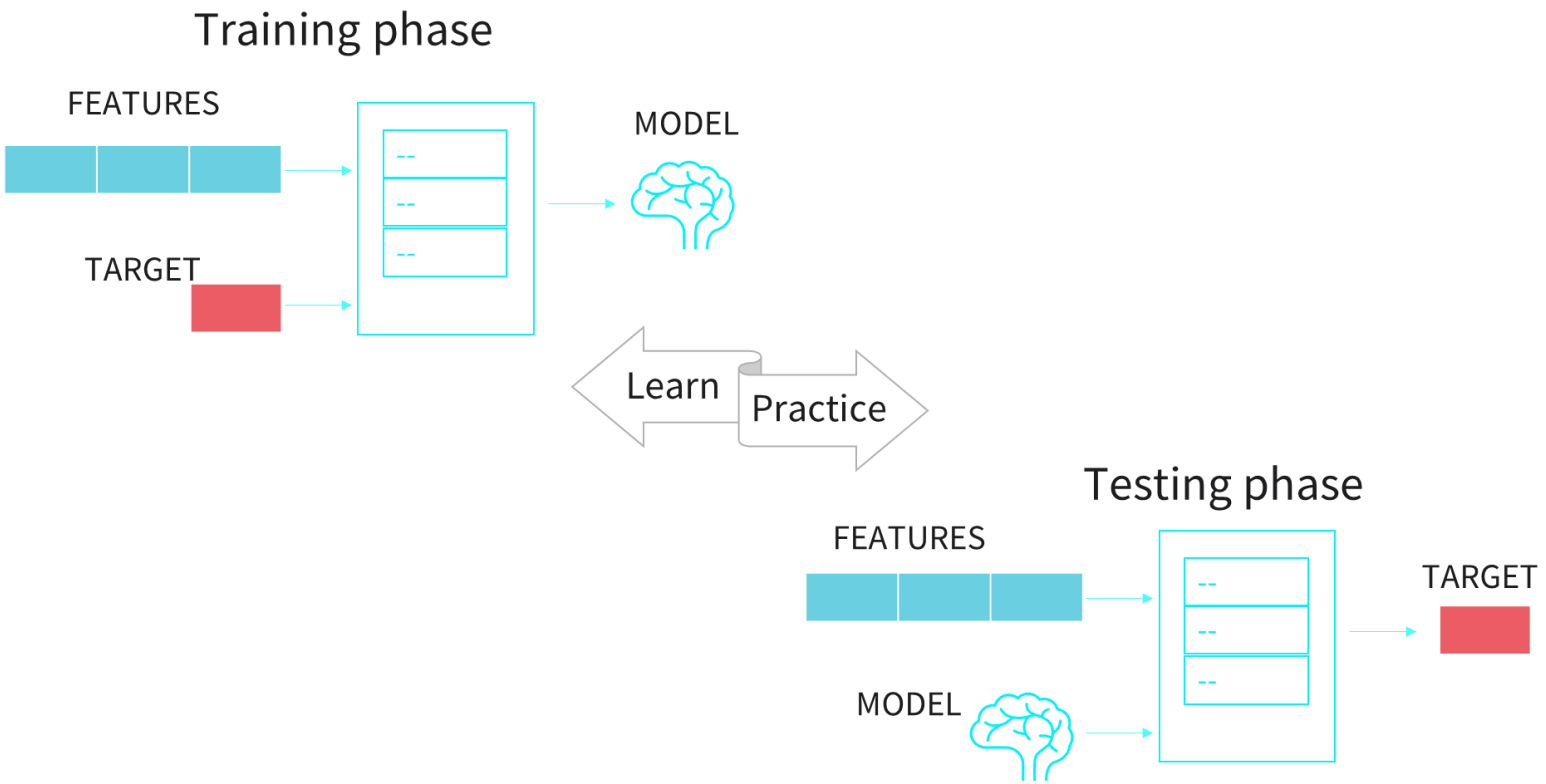
Why Machine Learning

- Computer systems learn and improve from experience



Categories of Machine Learning

Machine Learning Framework

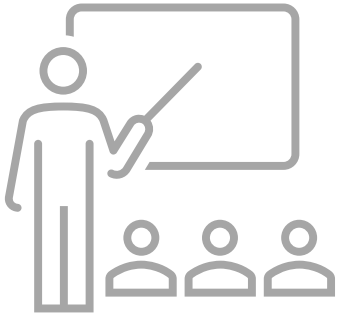


Supervised learning

- Supervised learning

  use labeled dataset to train

  correct output is known



- Applications

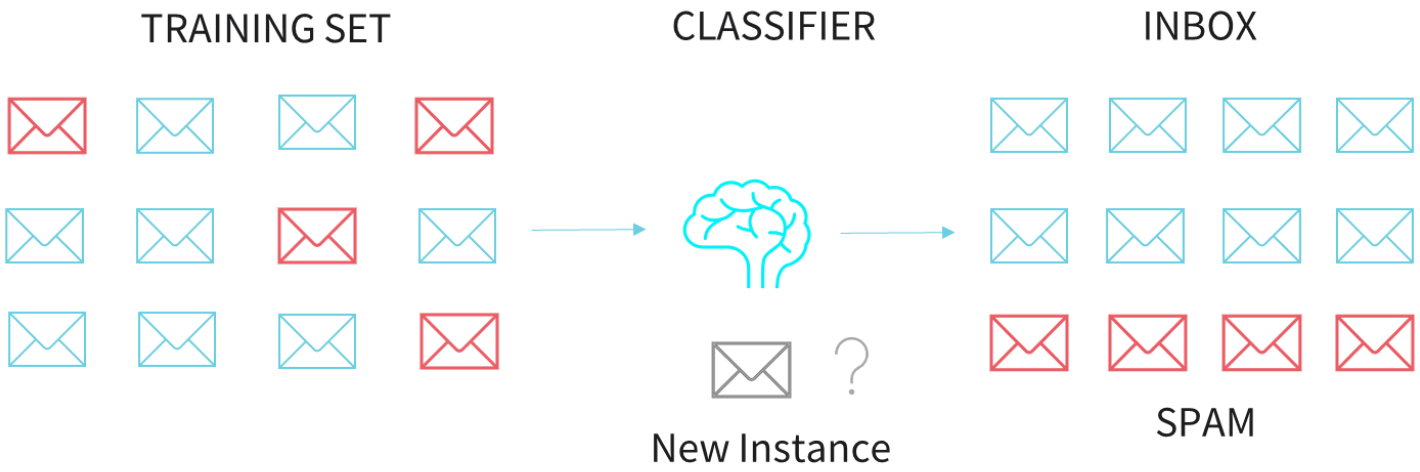
  image classification

  speech recognition

  weather forecast

Supervised Learning

- Classification example: filtering spam email

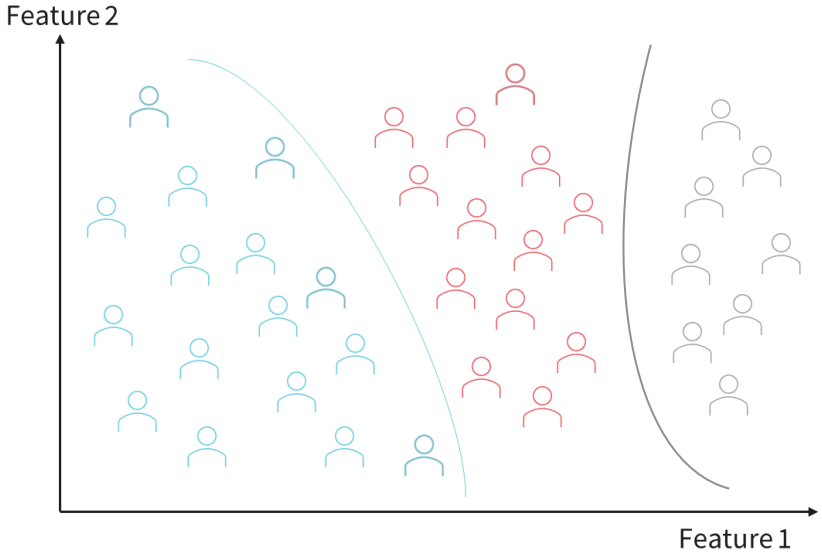


Supervised Learning

- Regression example: predicting house prices

Unsupervised learning

- Learn without a teacher
- Group set of related data
- Applications: data analytics, customer segmentation, recommender systems

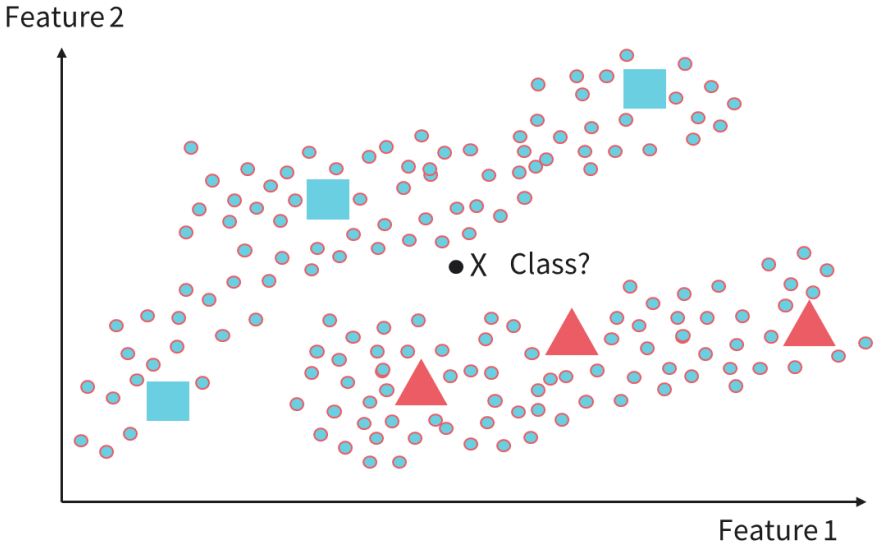


Semisupervised learning

- Tradeoff between supervised and unsupervised learning

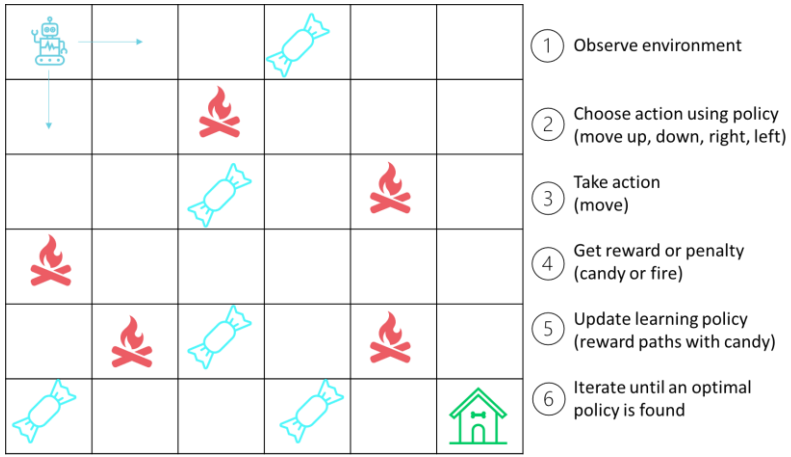
- Large set of unlabeled data for grouping

- Small set of labeled data for classification



Reinforcement learning

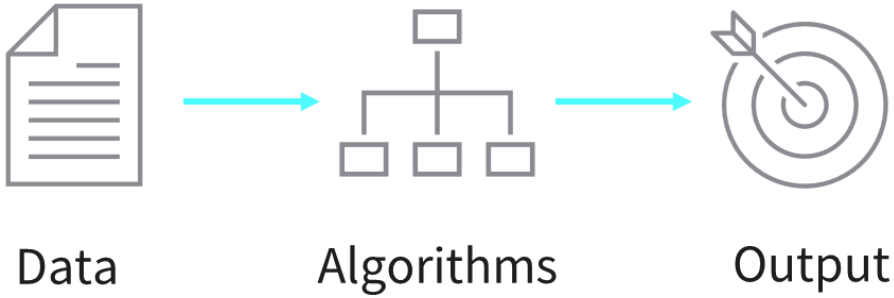
- Train through trial and error
- Reward desired behavior
- Punish undesired behavior



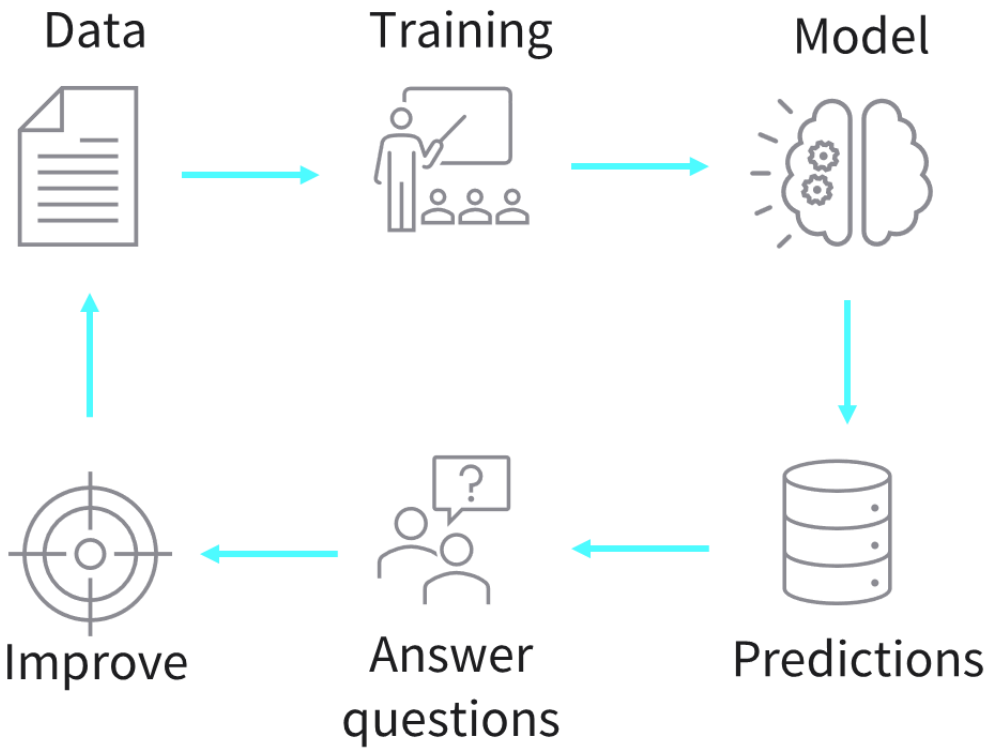
Fundamental Concepts

Fundamental concepts

- Machine learning uses **data** and **algorithms** to solve tasks



Fundamental concepts: the machine learning model



Any questions?